Studying grammatical change with the Penn Parsed Corpora of Historical English
Session 2: Practical: Starting out with CorpusSearch

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1. Introduction

This session will introduce you to CorpusSearch 2, a program designed to help you search corpora parsed in the Penn style: CorpusSearch is a simple program that only requires two things: a query file, and a source file (corpus).

1.1 Getting CorpusSearch set up

a. Go to http://walkden.space/uppsala.zip. Download the file.
b. Open the archive and unzip it in an easily accessible location (password: `argenteus`).

(You can also download CorpusSearch from https://corpussearch.sourceforge.net/.)

We'll be using a lemmatized version of the Penn Parsed Corpus of Early Modern English, created by Beatrice Santorini.

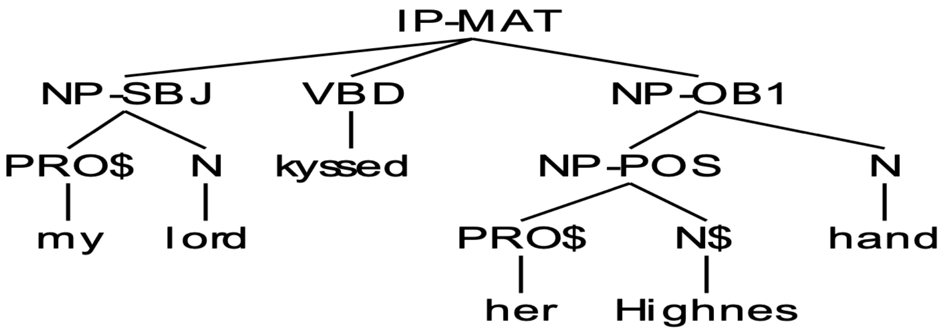
2. The Penn annotation scheme

2.1 Major category labels

- Phrase labels, which dominate another label, phrase or word-level
  - Clausal (CP, IP)
  - Non-clausal (NP, ADJP, ADVP, PP, etc.)
- Word labels, which dominate text
  - N, ADJ, ADV, P, etc
- Function is indicated on all IP-level NPs by an extended tag
  - -SBJ subject -OB1 first object (direct)
  - -ADV adverbial -OB2 second object (indirect)
  - -MSR measure -LOC locative
  - -TMP temporal -DIR directional
- Possessive NPs within an NP are labelled NP-POS
- IPs also have extended labels for functions:
  - -MAT main clause -SUB subordinate clause
  - -INF infinitival clause -INF-NCO non-complement infinitival clause
  - -SMC small clause -SPE direct speech (not mutually exclusive with the others)
- As do CPs:
  - -REL relative clause -FRL free relative clause
  - -THT *that*-clause -ADV adverbial clause
  - -CMP comparative -QUE interrogative
  - -DEG degree -SPE direct speech (not mutually exclusive with the others)
- For other and miscellaneous extended labels, see the manuals.

2.2 Parse represented as tree

```
(IP-MAT (NP-SBJ (PRO$ my) (N lord))
        (VBD kyssed)
        (NP-OB1 (NP-POS (PRO$ her) (N$ Highnes))
                (N hand))
```



2.3 Lemmatization

The lemmatized version has the **lemma** attached to the form of the word after a hash (#).

```
( (IP-MAT (ADVP-LOC (ADV there#there_adv))
     (BEP is#be_v)
     (NP-SBJ (D a#a_adj)
         (ADJP (ADV pretty#pretty_adv) (ADJ good#good_adj))
         (N house#house_n1)
         (PP (P of#of_prep)
              (NP (D the#the_adj)
                (N Lord#lord_n)
                (NP-PRN (NPR Lucas#NA)))))
     (PUNC :))
  (ID FIENNES-1698-E3-H,142.34))
```

2.4 Defining relations on a tree

- A node is a label in the tree (anything except text)
- Tree structures can be defined by a combination of three relations between nodes
    - Dominance: **X dominates Y if the subtree under X contains Y**
    - Sisterhood: **X and Y are sisters if they are immediately dominated by the same node**
    - Precedence: **X precedes Y if X is to the left of Y at any level in the tree**
- Most CorpusSearch functions are based on these three relations

2.5 Basic functions

- X Exists
    - X (Label or text) exists anywhere in token
- X hasSister Y
    - X and Y are sisters
- X Precedes Y
    - X precedes Y when X is to the left of Y
- X iPrecedes Y
    - X immediately precedes Y when X is to the left of Y and nothing intervenes between X and Y

- X iDominates Y (or: X iDoms Y)
    - X immediately dominates Y if X dominates Y and X and Y are exactly one generation apart
- X iDomsOnly Y
    - X immediately dominates Y as an only child
- X iDomsMod Y Z
    - X dominates Z and Y *may* occur in between X and Z
- A query file contains minimally two things
    - A *node* setting: The node sets the domain within which CorpusSearch searches
    - A *query*: Defines the structure being searched for

3. Getting started

3.1 A basic query

```
node: IP*
query: (NP-SBJ* iDoms PRO*)
```

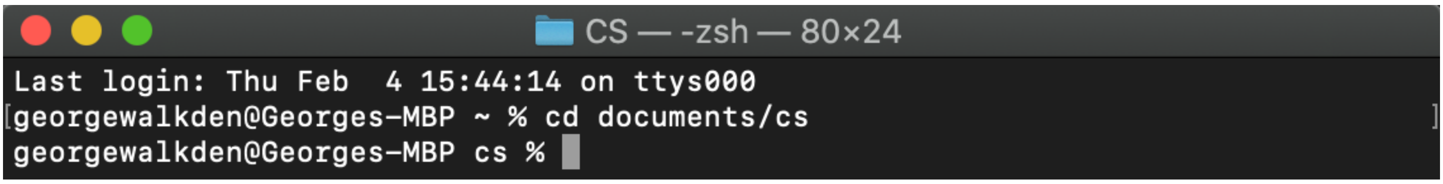
- **NP-SBJ*** and **PRO*** are search terms; search terms are labels, text, or definitions
- **iDoms** is a search function
- **(NP-SBJ* iDoms PRO*)** is a search function call

Open Notepad (Windows) or TextEdit (MacOS).
Type (or copy) the above basic query, then choose 'Save As' or 'Save…' and save it in your new 'CS' folder, in the same place where you . Give it a name of the form 'queryname.q'. Queries always have to end in **.q**.

- On Windows (Notepad) make sure you choose 'All files' rather than 'Text Documents' when you save the file.
- On MacOS (TextEdit) make sure you *uncheck* the box 'If no extension is provided, use ".txt".'

Now open the Command Prompt (Windows) or the Terminal (MacOS, under 'Applications' > 'Utilities').

- Within this window, change the working directory to the new CS folder you've created, using the command `cd` (e.g., in Terminal, `cd documents/cs`).



To load CorpusSearch, you need to type the following, and then press Enter:

- Windows: `java -classpath "CS_2.003.04.jar" csearch/CorpusSearch`
- MacOS: `java -classpath CS_2.003.04.jar csearch/CorpusSearch`

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(If this step doesn't work, you probably need to install Java and then try this again.)

```
CS — java -classpath CS.jar csearch/CorpusSearch — 80×24
Last login: Thu Feb  4 15:44:14 on ttys000
georgewalkden@Georges-MBP ~ % cd documents/cs
georgewalkden@Georges-MBP cs % java -classpath CS.jar csearch/CorpusSearch

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CorpusSearch version 2.003.00

CorpusSearch Manual:  http://corpussearch.sourceforge.net/CS-manual/Contents.htm
l

Enter name of query file (q to quit, ? for help):
```

- Our query is the one you created, `queryname.q`. Type that in and press Enter.
- Next it'll ask you for a source file. We'll search the lemmatized PPCEME. Type in `PPCEME-24/*.lem` then press Enter.
- It'll suggest a name for your output file. Press `y` then Enter. If it says that the file already exists, press `y` then Enter again.
- Now it will start searching. The below window is what it looks like when finished.

```
CS — -zsh — 80×24
Last login: Thu Feb  4 15:44:14 on ttys000
georgewalkden@Georges-MBP ~ % cd documents/cs
georgewalkden@Georges-MBP cs % java -classpath CS.jar csearch/CorpusSearch

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CorpusSearch version 2.003.00

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l

Enter name of query file (q to quit, ? for help):  firstquery.q
Enter name(s) of source file(s), q to quit:  shakesp-e2-p2.psd
Would you like to name your output file firstquery.out (Y/N)?  y
File firstquery.out already exists.  Overwrite?  (Y/N):  y
Working.  Please be patient.
Output file is
    firstquery.out
time taken:  72651 milliseconds.  1 minutes, 12 seconds.



Search completed.
georgewalkden@Georges-MBP cs %
```

3.2 Understanding the output

The output of the search – e.g. all tokens matching the description you specified in the query file – is saved in a **.out** file, by default with the same name and in the same folder as the original query. Load Notepad or TextEdit again, and open the **.out** file (‘**queryname.out**’). The file starts with information about your query:

```
/*
PREFACE:
Copyright 2010 Beth Randall
Date:  Sat Jun 06 21:11:47 CEST 2026

command file:     queryname.q
input files:      PPCEME-24/abott-1530-e1-p1.lem
through:          PPCEME-24/zouch-1649-e3-p2.lem
448 files.
output file:      corpus_stuff/queryname.out

node:   IP*
query:  (NP-SBJ* iDoms  PRO*)
*/
```

After this, all the examples are given, separated by a header and footer.

```
/*
HEADER:
source file:  abott-1530-e1-p1.lem
*/
```

Here’s one example:

```
  /~*
  It#it_pron was#be_v my#my_adj consent#consent_n;
  (ABOTT-1530-E1-P1,229.7)
  *~/
  /*
  1 IP-MAT:  2 NP-SBJ, 3 PRO
  */

  ( (IP-MAT (NP-SBJ (PRO It#it_pron))
        (BED was#be_v)
        (NP-PRD (PRO$ my#my_adj) (N consent#consent_n))
        (PUNC ;))
    (ID ABOTT-1530-E1-P1,229.7))
```

After each file is a footer.

```
/*
FOOTER
  source file, hits/tokens/total
  abott-1530-e1-p1.lem 30/19/29
*/
```

At the bottom of the whole file, a summary contains information about the search overall.

```
/*
SUMMARY:
source files, hits/tokens/total
  abott-1530-e1-p1.lem  30/19/29
  alhatton-1700-e3-h.lem      57/34/48
  alhatton2-1687-e3-p1.lem    65/36/74
  …
whole search, hits/tokens/total
      96144/58450/105513
*/
```

- **Hits:** number of distinct boundary nodes (in this case, IP*) containing the searched-for structure.
- **Tokens:** number of independent parsed objects (tokens) in which hits occurred. This can be lower than Hits if there is more than one hit per token (in this case, more than one NP-SBJ* immediately dominating a PRO*).
- **Total:** total number of independent parsed objects (tokens) searched.

4. Logical operators

- CorpusSearch uses the logical operators 'not' and 'or' in conjunction with search terms
  - 'not' is represented by !
    - !NP-OB1 = "not NP-OB1"
    - e.g. `(IP* iDoms !NP-OB1)`
  - 'or' is represented by |
    - Join a list of labels with |
    - e.g. `(IP* iDoms NP-SBJ|NP-OB1|NP-OB2)`

5. An important fact

* is a wildcard! It stands for any number of arbitrary characters (including none).

So VB* will match VB, VBP, VBD, etc.

*consider** will match *consider*, *considering*, *considered*, *consideration*, etc.

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6. Exercises

The manual for the corpora is at https://www.ling.upenn.edu/~beatrice/corpus-ling/annotation-2022/

Using that manual, and repeating most of the steps given above, keeping the node the same and using the same corpus (PPCEME-24), can you find the following:

- Present tense lexical verbs (VBP) immediately preceding negation (NEG)?
  - Query: ________________________
  - Number of hits: ________________________
- A present tense modal (MDP) or form of 'be' (BEP) followed by an adverb (ADV)?
  - Query: ________________________
  - Number of hits: ________________________
- The word 'Doctor' used as a proper noun (NPR)?
  (Hint: part-of-speech tags immediately dominate the word they tag.)
  - Query: ________________________
  - Number of hits: ________________________
- A preposition with a noun phrase for a sister?
  - Query: ________________________
  - Number of hits: ________________________
- A preposition phrase immediately dominating only a preposition?
  - Query: ________________________
  - Number of hits: ________________________

Acknowledgements

Thanks to Prof. Ann Taylor (University of York), on whose workshop presentation some of this handout is closely based. For more detail on CorpusSearch 2, the manual can be found at https://corpussearch.sourceforge.net/CS-manual/Contents.html.

Appendix: Searching the CorpusSearch way: an algorithm for effective searching
(thanks again to Prof. Ann Taylor, University of York, for this appendix)

1. Choose an appropriate node

- First, consider what you want your search domain to be
    - Which node you select will affect which nodes are removed
    - Which node you select will affect how CorpusSearch counts

2. Tie your search terms to the node

- If you are writing a multi-call query:
    - Use the node as your first search term
    - Tie the node to a subsequent search term using iDoms
    - Tie subsequent search terms to each other using iDoms or hasSister

3. Search incrementally

- CorpusSearch was designed to search on its own output, so make use of this feature
- Always keep a (written) record of your sequence of searches and what each one is meant to accomplish, so that you can repeat part or all of the sequence if necessary

4. Check your data

- The annotation system is complicated and it is easy to make a mistake
- Check your data at every possible point to make sure you are getting what you want
- Always start by searching widely (use * a lot) and then narrow down the search
- It is easier to recognise unwanted data than to realise that something is missing

5. Don't worry, be happy

- CorpusSearch searches very quickly
    - If you make a mistake writing the query, it's no big deal, just try again
    - And again, and again...
    - Even if you have to do 10 (or 20, or 30) searches to get what you want, the time invested is still very small

6. Experiment!

- One of the best ways to find out how the corpus is annotated is to search it
    - When you're starting out, try a general search on the category you're interested in (e.g., CP-REL* exists) and see what you get
    - Then refine the searches as you figure out more about how the structures you want are annotated
- But don't forget to read the relevant parts of the manual as well!

Answers to exercises

- Present tense lexical verbs (VBP) immediately preceding negation (NEG)? `(VBP iPrecedes NEG)`
- A present tense modal (MDP) or form of ‘be’ (BEP) followed by an adverb (ADV)? `(MDP|BEP Precedes ADV)`
    - (NB: remember the difference between Precedes and iPrecedes!)
- The word ‘Doctor’ used as a proper noun (NPR)? `(NPR iDoms Doctor#*)`
- A preposition with a noun phrase for a sister? `(P hasSister NP)`
- A preposition phrase immediately dominating only a preposition? `(PP iDomsOnly P)`
    - There are no PPs immediately dominating only Ps.)

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